

PRODUCT BRIEF AND MVP PLAN

Mu'allim

المعلم

AI fault diagnosis for maintenance teams,
built on each company's own repair history



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Prepared during the 'Build a Product with AI' workshop

Covers: product, customers, features, technology, business model, go-to-market,
8-week MVP and 4-week pilot plan, risks

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SECTION 01

Executive summary

Mu'allim (Arabic for the master technician) puts a company's best technician in every technician's pocket. A technician scans the QR sticker on a machine, describes or photographs the fault, and gets the most likely causes, the proven fix, the parts, and the safety steps. Every answer comes from the company's own repair history and manuals, with the evidence shown. Every confirmed repair makes the next answer better.

What the customer is buying

Faster repairs Less downtime per breakdown, because diagnosis starts from what already worked on this exact machine model.	Knowledge that stays When a senior technician leaves or retires, their experience stays in the system as structured, searchable knowledge.	Visibility Managers finally see which machines fail, why, how often, and how long repairs take.
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At a glance

Topic	Summary
Customer (buyer)	Maintenance and plant managers in factories with 50 to 500 employees, and maintenance service contractors
Daily users	Shop-floor technicians; maintenance engineers review and approve knowledge
Starting point	One machine family, chosen together with a design partner who shares real fault data
Target experience	The right cause in the top 3, with evidence, in under 30 seconds from scanning the machine
Moat	Each customer's structured fault history plus a feedback loop that improves with every repair
Business model	Paid pilot, then subscription per site and technician seat, plus a one-time onboarding fee
Plan	8 weeks to build the MVP, then a 4-week pilot with 5 to 10 technicians at one site

SECTION 02

The problem

When a machine stops, every hour costs production, overtime, and delivery delays. Yet in most maintenance teams, diagnosis still depends on a few experienced people and scattered records:

- the memory of one or two senior technicians
- paper or PDF manuals that nobody can search quickly on the shop floor
- fault logs in Excel, maintenance software, or WhatsApp photos, written differently by each person
- calls to the machine supplier's service line, which can take hours or days

The result: the same faults are diagnosed from scratch again and again, junior technicians take much longer, and knowledge walks out of the door when experienced people leave.

What teams do today, and where it breaks

What teams do today	Where it breaks
Call the senior technician	They are busy, off shift, or have left the company
Search the manual	Slow; manuals describe the machine, not this plant's real faults
Fault log in Excel or maintenance software	Inconsistent records, hard to search by symptom
WhatsApp group with photos	Answers get lost; nothing is structured or reusable
Supplier hotline or technician visit	Expensive and slow for common faults
General AI chatbot	Doesn't know this machine's history and may guess confidently

The opportunity: the knowledge already exists inside the company. It is scattered, unstructured, and in people's heads. Mu'allim turns it into answers a technician can use in seconds.

SECTION 03

Customers and starting market

Three roles, three needs

Buyer: maintenance manager

- Owns downtime and the budget
- Wants fewer emergencies and proof for management
- Signs the pilot and the contract

Daily user: technician

- Works with gloves, noise, and weak signal
- Wants a fast answer they can trust
- Must find it faster than calling a colleague

Knowledge owner: engineer

- Approves knowledge, handles escalations
- Wants fewer repeated questions
- Wants a clean record of what works

Start with one machine family

A narrow start gives better answers faster. Choose a family that is common across many plants, where faults repeat, downtime is expensive, and a design partner will share real records. **Data access decides the choice.**

Machine family	Why it can work	Watch out for
Industrial air compressors	In almost every factory; common, well-documented fault types	Many brands and models; start with one or two
Packaging machines	Frequent stops with high cost per stop on production lines	Faults are often model-specific
CNC machines	Clear error codes on the controller; expensive downtime	Codes depend on the controller brand
Generators	Critical equipment; contractors maintain many units	Few faults per single site; better through contractors
Chillers and HVAC	Many units per site; service contractors	Mix of electrical and refrigeration faults

First customer types

- 1 **Factories with 50 to 500 employees** and a small maintenance team: short decision path, real pain.
- 2 **Maintenance service contractors:** one contract covers many machines of the same type, which means more data and faster learning.
- 3 **Machine distributors and manufacturers** (later): they can offer the product to all of their customers as an added service.

SECTION 04

How it works: the core loop

The whole product is built around one loop that runs on every breakdown:



Step	What happens
1 Report	The technician scans the machine's QR sticker, then describes the problem by text or voice, takes a photo, or enters an error code.
2 Diagnose	The system shows the top likely causes with a confidence level and the past cases and manual pages each one is based on.
3 Fix	A step-by-step checklist with safety warnings first, then parts, tools, and estimated time.
4 Confirm	The technician logs fixed, partially fixed, or not fixed, the actual cause, the time spent, and photos.
5 Learn	Confirmed fixes strengthen the knowledge base. New faults become draft records for the engineer to approve.

Why the loop matters: after months of use, the system holds a structured record of how this company's machines fail and how they get fixed. A competitor can copy features, but not that history. This is the moat.

SECTION 05

Product features by module

5.1 Knowledge ingestion

The hardest part, and the most valuable one.

- Import fault catalogs from Excel or CSV, PDF manuals, and exports from maintenance software.
- AI turns messy records into structured fault records: machine model, component, symptoms, error codes, root cause, fix steps, parts, tools, estimated time, and safety notes.
- Duplicate detection: the same fault written five different ways becomes one record.
- Review queue: a maintenance engineer approves every AI-structured record before it is used in answers.
- Version history for every record.

MVP: Excel/CSV, PDF, AI structuring, review queue. Later: maintenance software exports, duplicate detection, continuous sync.

5.2 Asset registry

Knowing the exact machine makes answers far more accurate.

- Hierarchy of sites, lines, and machines, each with model, serial number, and install date.
- A printed QR sticker per machine. Scanning it opens a diagnosis with the right machine selected.
- Machine history: every past fault and repair on that exact machine.

MVP: machine list and QR stickers. Later: full hierarchy and history timeline.

5.3 Diagnosis assistant

The screen technicians use every day.

- Inputs: text or voice in Arabic and English, photos of error displays, damaged parts or leaks, and error codes.
- Guided questions when the input is unclear, the way an experienced technician narrows things down ("Is the pressure above or below normal?").
- Output: the top 3 likely causes with a confidence level, each linked to the past cases and manual pages it is based on; fix steps, parts, tools, and time; safety warnings shown first.
- No guessing: when nothing similar exists, it says so and offers escalation to a senior technician.

MVP: text, photo, error code, top 3 with evidence, escalation. Later: Arabic voice, guided questions.

5.4 Repair execution and feedback

Turns every repair into better knowledge.

- Step checklist the technician ticks off, with before and after photos.
- Outcome: fixed, partially fixed, or not fixed, plus the actual cause and time spent.
- New fault capture: unknown faults become draft records for the engineer to review.

MVP: outcome, actual cause, new fault drafts. Later: step checklist with photos.

5.5 Manager dashboard

Makes the value visible to the person who pays.

- Average repair time and how it changes over time.
- Top failure causes per machine and line, and repeat faults on the same machine.
- Knowledge coverage: which machines have good data and which don't.
- Usage and logged outcomes per technician; export to Excel or PDF.

MVP: faults per machine, outcomes, usage. Later: trends, repeat faults, reports.

5.6 Enterprise foundations

What makes it acceptable to a company.

- Complete data separation between customer companies.
- Roles: technician, engineer, manager, and admin.
- Audit log of who approved which knowledge and who did which repair.
- Choice of data region; later, on-premise deployment with self-hosted models for plants that do not allow cloud.

MVP: data separation and roles. Later: audit log, company sign-in, on-premise.

5.7 Later: audio, integrations, prediction

Built once the core loop works.

- Audio: record each machine's healthy sound as a baseline, detect differences, and match new sounds to recordings attached to past cases.
- Integrations: work orders in the company's maintenance software (CMMS), spare-parts stock, and alerts on WhatsApp or Microsoft Teams.
- Offline mode for areas with weak signal.
- Sensor data for early warnings before failures happen.

V1 and V2.

A note on audio. General AI models are weak at diagnosing machine noise from a single recording. Audio becomes reliable when a new sound is compared with a baseline from the same machine, or with recordings attached to past cases. That is why it comes after photos, text, and error codes.

SECTION 06

What goes into each version

The MVP proves the core loop at one site. V1 makes it comfortable for daily use; V2 makes it sticky inside the customer's systems.

Module	MVP (8 weeks)	V1	V2
Ingestion	Excel/CSV and PDF manuals, AI structuring, review queue	Maintenance software exports, duplicate detection	Continuous sync
Assets	Machine list, QR stickers	Full site and line hierarchy	Machine history timeline
Diagnosis	Text, photo, error code; top 3 with evidence; safety; escalation	Arabic voice input, guided questions	Audio baselines
Feedback	Outcome, actual cause, new fault drafts	Step checklist with photos	Automatic knowledge improvements
Dashboard	Faults per machine, outcomes, usage	Repair time trends, repeat faults	Reports and exports
Enterprise	Data separation, roles	Audit log, company sign-in	On-premise deployment
Integrations	None	Work orders, WhatsApp alerts	Spare parts, sensors

SECTION 07

The diagnosis engine

Each diagnosis goes through seven steps. The design goal is simple: an answer is only as good as the evidence behind it, so the engine never answers from general knowledge alone.



- 1 **Context.** The QR scan provides the machine model, its history, and its documents.
- 2 **Understand the input.** Voice is transcribed; a vision model reads error codes from displays and describes visible symptoms in photos.
- 3 **Search three ways.** Exact match on error codes, keyword search on symptoms, and semantic search on the meaning of the description, all filtered by machine model, then re-ranked for relevance.
- 4 **Rank causes.** The model ranks causes using only the retrieved cases and manual pages, and must cite them.
- 5 **Confidence gate.** If the best match is weak, the system says "not enough data" and offers escalation instead of guessing.
- 6 **Safety check.** Every fix includes its required safety steps; steps that need a qualified person are flagged.
- 7 **Log everything.** Inputs, evidence, and the answer are stored for the feedback loop and for measuring quality.

Design principles

Grounded

Answers come only from this customer's approved knowledge and manuals.

Evidence shown

Every cause links to the past cases and manual pages behind it.

Honest

When the data is not there, the system says so and escalates.

Safety first

Safety warnings appear before any fix step.

Human decides

The technician stays responsible; the system advises.

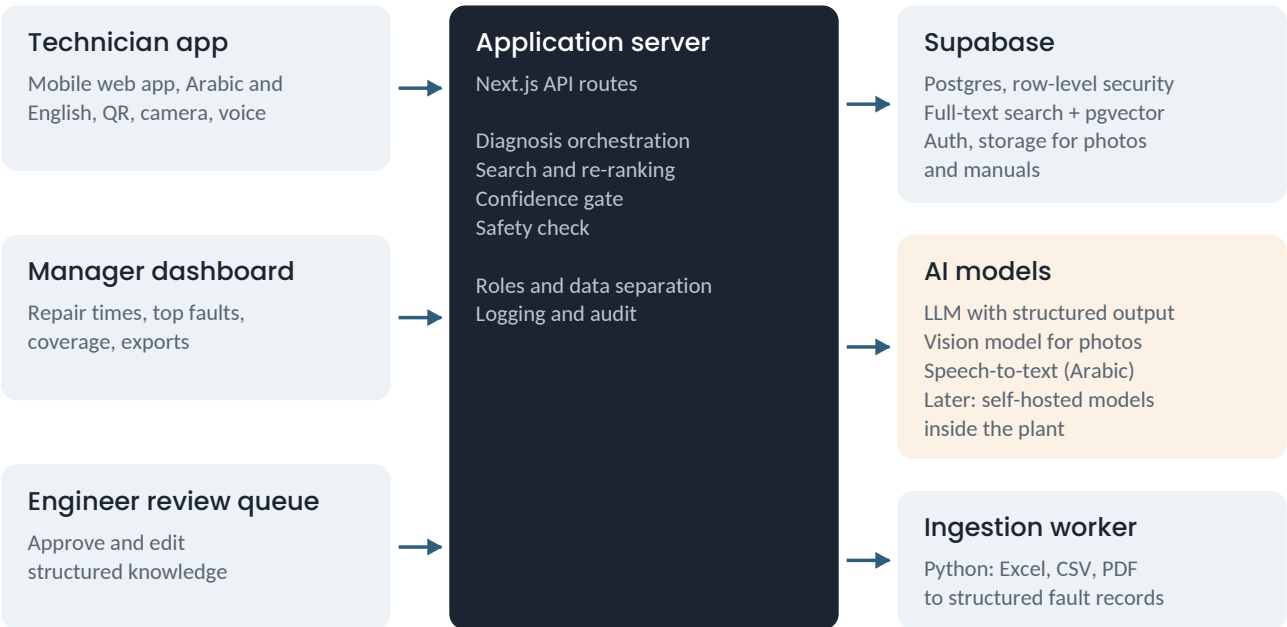
Fast

QR scan plus one photo should give an answer in under 30 seconds.

SECTION 08

Architecture and technology

A deliberately simple stack for the MVP, chosen so that one or two developers can build it in eight weeks and later move it on-premise for customers who need that.



Layer	Choice	Notes
Frontend	Next.js, mobile-first web app	Installable on phones; Arabic and English with right-to-left support; large touch targets for gloved hands
Backend	Supabase	Postgres database, authentication, file storage for photos and PDFs
Search	Postgres full-text search + pgvector	Keyword and semantic search combined, then re-ranked
Ingestion	Python worker	Parses Excel and PDF files; uses an LLM to extract structured records
AI models	LLM, vision model, speech-to-text	Choose models by results on the test set; strict structured output for extraction
Security	Row-level security per company	Encrypted storage, audit log, least-privilege access
On-premise (later)	Self-hosted open-weight models	Served inside the customer network, for example with vLLM

Core data model

Entity	Purpose	Key fields
companies	Customer accounts	name, region, plan
users	People and roles	company, role (technician, engineer, manager, admin), language
sites, lines, machines	Asset registry	model, serial number, install date, QR code
machine_models	Shared per model	manufacturer, model name, documents
fault_records	The knowledge base	symptoms, error codes, root cause, fix steps, parts, tools, time, safety notes, status (draft or approved), version
documents, chunks	Manuals and imported files	source file, page, text, embedding
diagnoses	Every request	machine, inputs, photos, retrieved evidence, ranked causes, confidence
repair_outcomes	Feedback	diagnosis, result, actual cause, time spent, photos, notes
audit_log	Traceability	actor, action, target, timestamp

SECTION 09

Measuring AI quality

Before any technician sees the product, build a **test set**: 50 to 100 real past faults from the design partner where the actual cause is known. Hide the answers, run them through the system, and score the results. Re-run the test set after every change to prompts, search, or models.

Metric	What it tells you
Top-3 accuracy	The real cause appears among the top 3 suggestions
Top-1 accuracy	The real cause is ranked first
Correct "not enough data"	When the system declined to answer, the knowledge really was missing
Evidence correctness	The cited cases and manual pages actually support the suggested cause
Safety coverage	Every fix that needs safety steps shows them
Response time	Time from submitting the report to seeing the answer

Set targets with the design partner before the pilot, for example the share of cases where the real cause must be in the top 3. Agreeing on this upfront turns the pilot into a clear yes or no decision instead of a matter of opinion.

SECTION 10

Business model and pricing

Component	How it works	Why
Paid pilot	2 to 3 months at a reduced price	Proves real interest; free pilots are easier to get but send a weaker signal
Subscription	Per site plus per technician seat, or per machine	Grows as the customer adds lines, sites, and technicians
Onboarding fee	One-time	Covers importing and structuring existing data, the most labor-intensive step
On-premise (later)	Annual license plus setup	For plants that cannot send data to the cloud

ROI worksheet for sales conversations

Fill this in with the customer's own numbers during the first meeting. Their numbers are far more convincing than yours.

	Input	Value
A	Breakdowns per month on the covered machines	_____
B	Hours saved per breakdown (measured in the pilot)	_____
C	Cost of one hour of downtime (from the customer)	_____
D	Monthly value = $A \times B \times C$	_____
E	Monthly price of the subscription	_____

The tool pays for itself when D is greater than E. Also count the value that is harder to put in numbers: less dependence on one or two people, faster onboarding of new technicians, and fewer supplier callouts.

SECTION 11

Go-to-market

Channels, in order

1 Design partners

- 1 to 2 companies that share data
- A maintenance manager who wants this
- Shape the product and become references

2 Maintenance contractors

- One contract, many machines and sites
- Same machine types repeat
- Faster learning per customer

3 Distributors and makers

- Offer it to all their customers
- Many customers through one partner
- Natural fit after V1

Sales process for each customer

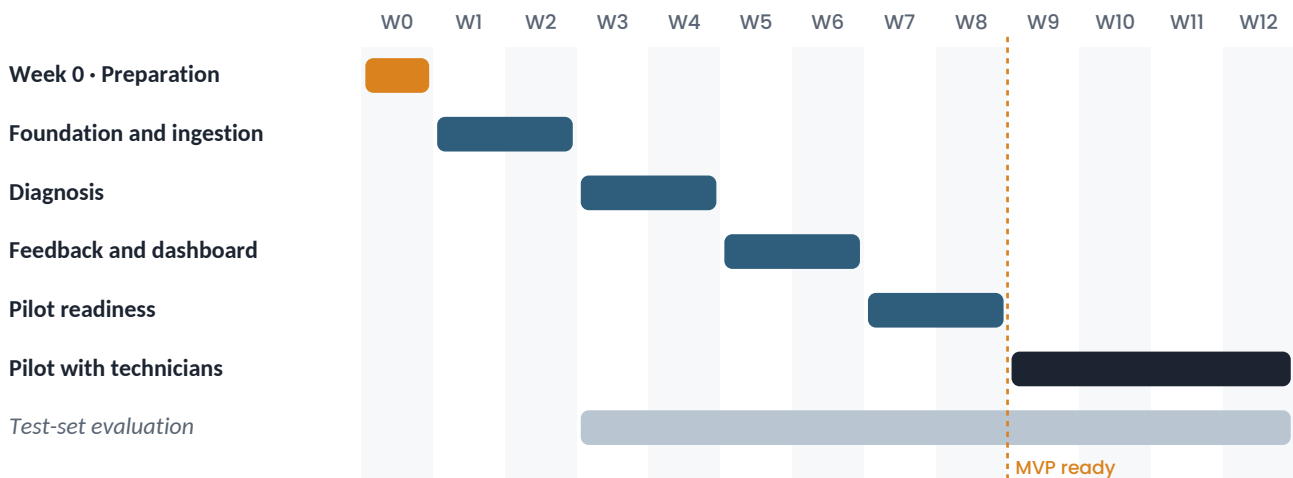
Step	What happens	Output
1 Discovery	Meet the maintenance manager: pain, downtime cost, what data exists	Fit or no fit; filled ROI inputs
2 Data audit	Review a sample of 20 to 50 past repair records and one manual	Data quality assessment
3 Test on their data	Import the sample and run their own past faults through the system	Top-3 accuracy on their cases
4 Paid pilot	One site, one machine family, agreed success metrics	Measured results
5 Rollout	More lines, sites, and machine families	Subscription contract

Where to start: Jordan's industrial estates, such as Sahab, through personal networks and introductions. Once the product works for one machine family in Jordan, the Gulf is the natural next market.

SECTION 12

MVP plan: 8 weeks plus a pilot

The plan runs in two-week phases. Each phase ends with something the design partner can see and react to.



Phase	Deliverables	Done when
Week 0 Preparation	- Choose the machine family with the design partner - Collect the fault catalog, 2 to 5 manuals, and 50 to 100 past repair records - Build the test set - Agree on pilot success criteria	Data received and a test set of at least 50 cases with known causes
Weeks 1-2 Foundation and ingestion	- Company accounts, roles, data separation - Machine list and QR codes - Import Excel/CSV and PDF manuals - AI structuring and the review queue	The partner's catalog is imported and the first records are approved by their engineer
Weeks 3-4 Diagnosis	- Technician app: QR, text, photo, error code - Combined search and re-ranking - Answers with evidence and safety warnings - Confidence gate and escalation - First test-set run	Answers come back with evidence and the first test-set score is recorded
Weeks 5-6 Feedback and dashboard	- Outcome, actual cause, new fault drafts - Basic manager dashboard - Accuracy improvements from test-set results	The full loop works end to end and the test-set score has improved
Weeks 7-8 Pilot readiness	- Arabic voice input - Security review, speed, error handling - Onboarding guide and a 30-minute technician training - QR stickers printed and attached	The pilot site is ready and technicians are trained
Weeks 9-12 Pilot	5 to 10 technicians use it on real breakdowns - Weekly review with the maintenance manager	Go, fix, or stop decision with the manager

SECTION 13

Pilot plan and success metrics

Setup: one site, one machine family, 5 to 10 technicians, 4 weeks, and a weekly 30-minute review with the maintenance manager.

Area	Metric	How it is measured
Adoption	Diagnoses per technician per week	App logs
Discipline	Share of diagnoses with an outcome logged	App logs
Accuracy in the field	Share of repairs where the actual cause was in the top 3	Outcome records
Speed	Time from report to diagnosis	App logs
Impact	Repair time compared with before the pilot	Pilot records vs. historical records
Knowledge growth	New fault records approved	Review queue
Willingness to pay	The manager wants to continue on a paid plan	End-of-pilot meeting

Decision at the end of the pilot

Go

Targets met and the manager wants to continue: convert to a subscription and expand to more lines or sites.

Fix

One clear gap, such as data coverage or speed: fix it and extend the pilot by 2 to 4 weeks.

Stop or pivot

Technicians did not use it or accuracy stayed low: review the machine family, the data, or the customer type.

SECTION 14

Team and running costs

Team for the MVP

Role	Responsibility	Time
Full-stack developer	Technician app, dashboard, backend, deployment	Full time
AI engineer	Ingestion, search, prompts, test-set evaluation	Full time (can be the same person in the first weeks)
Domain expert	A maintenance engineer from the design partner who reviews knowledge and answers	A few hours per week
Product and sales lead	Customer meetings, pilot management, pricing	Part time to full time

Running costs to plan for

- AI model usage per diagnosis and per imported document (LLM, vision, speech-to-text)
- Hosting and database
- File storage for photos and manuals
- QR sticker printing and installation
- Travel to the pilot site

Track cost per diagnosis from day one. It is the number that decides whether the subscription price leaves a healthy margin as usage grows.

SECTION 15

Risks and how to handle them

Risk	Impact	How to handle it
Poor data quality	Weak or wrong answers	Review queue; start with the cleanest data; the feedback loop improves records over time
Wrong or unsafe advice	Damage, injury, liability	Evidence-linked answers, confidence gate, safety checks; the technician stays responsible; terms reviewed by a lawyer
Technicians don't use it	No data, no value	Make it faster than calling a colleague; involve technicians in design; QR plus one photo gives an answer in under 30 seconds
Slow B2B sales	Cash runs out	Paid pilots with measurable results; contractors and distributors as channels
Data security concerns	Deals blocked by IT	Strict data separation, audit logs, region choice, on-premise option
AI model costs	Margins shrink	Track cost per diagnosis; cache results; use smaller models where they perform well enough
Only one design partner	Product fits one company only	Add a second partner before V1

SECTION 16

The one-week workshop version

For the workshop competition, a condensed version is enough to prove the idea:

- Import one sample fault catalog and one manual for a single machine type
- A web app where a technician describes or photographs a fault
- Top 3 causes with evidence and safety notes
- A "did it work?" button that saves the outcome

Use a realistic sample catalog, for example 50 air-compressor faults, and label it clearly as sample data.

2-minute demo script

- 1 Show the messy Excel catalog, then the structured records after import.
- 2 Select the machine, or scan its QR code.
- 3 Describe the fault in Arabic, or upload a photo of the error display.
- 4 Show the top 3 causes, each with its evidence and the safety warning.
- 5 Press "Fixed" and show the outcome appear on the dashboard.

SECTION 17

Next steps this week

- ☐ Pick three target companies and book discovery meetings
- ☐ Ask each for a sample of 20 to 50 past repair records and one manual
- ☐ Decide the machine family together with the first design partner
- ☐ Build the test set from their records
- ☐ Set up the repository and start Week 1 of the MVP plan

APPENDIX A

Example fault record

How one structured record looks after ingestion and engineer approval. *Illustrative example with sample content.*

Field	Example
Machine model	Rotary screw air compressor, [manufacturer and model]
Component	Oil cooler
Symptoms	High discharge temperature alarm; the unit trips after running under load for a while
Error code	[controller temperature alarm code]
Root cause	Oil cooler fins blocked with dust, reducing cooling
Fix steps	1. Stop and isolate the compressor (lockout/tagout). 2. Release pressure and let the unit cool down. 3. Clean the cooler fins following the manual procedure. 4. Check oil level and room ventilation. 5. Restart and monitor temperature under load.
Parts	None in most cases; cleaning agent if needed
Tools	Compressed air, soft brush, personal protective equipment
Estimated time	[from past repairs]
Safety notes	Lockout/tagout; release stored pressure; hot surfaces; hearing protection
Evidence	Fault log rows [ids]; manual page [number]
Status	Approved by [engineer], version 3

APPENDIX B

Example diagnosis screen

What the technician sees after scanning the machine and describing the fault. *Illustrative example with sample data.*

COMPRESSOR C-02 · LINE 2

"Temperature alarm, stops after half an hour"

SAFETY FIRST

Lock out and tag out, release pressure, and let the unit cool before opening. Hot surfaces.

1 Blocked oil cooler

High confidence · 6 similar past cases on this model; manual p. [38]

2 Low oil level

Medium confidence · 3 past cases; manual p. [22]

3 Faulty thermostatic valve

Low confidence · 1 past case

Start fix

Not this

Escalate

How to read it

- **Machine context** comes from the QR scan.
- **Safety** always appears before any cause or step.
- **Confidence** reflects how strong the matching evidence is.
- **Evidence** links to the exact past cases and manual pages.
- **Escalate** sends the case to a senior technician with everything attached.
- **Not this** records a wrong suggestion, which also improves future answers.

APPENDIX C

Discovery questions

Use these in the first meeting with a maintenance manager. Ask about what happened, not about what they would do in the future.

- 1 Which machines stop most often, and what does one hour of downtime cost you?
- 2 Walk me through the last breakdown: who diagnosed it, how, and how long did it take?
- 3 Where do you record faults and repairs today? Can we look at a sample?
- 4 What happens when your most experienced technician is not available?
- 5 How do new technicians learn the machines?
- 6 Have you tried any software for this? What worked and what did not?
- 7 Who would need to approve a pilot, and what would success look like for you?
- 8 Are there rules about data leaving the plant?